# VANISHING OF MULTIPLICATION ON THE COTANGENT COMPLEX PERSISTS UNDER BASE CHANGE

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The counted proof is the author's complete argument: the counted statement is the author lemma environment labelled lemma-zero-after-modding-out; the proof environment immediately following it is its own proof

*Editorial note.* The label lemma-zero-after-modding-out names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

*Editorial note.* The counted statement is carried by the author's own lemma environment in the chapter source.

*Allowed standard prerequisites (editorial).*

- the base change setup for pairs (A1, I1) to (A2, I2) of Noetherian adic rings (remark-base-change) with B1 in the completion class (equation-C-prime) and B2 its base change
- the base change map for naive cotangent complexes (lemma-NL-base-change): a map NL_{B1/A1} tensor B2 B1 to NL_{B2/A2} inducing an isomorphism on H^0 and a surjection on H^{-1}
- the More on Algebra lemma transferring vanishing of a self-map of a two-term complex under a map that is an isomorphism on H^0 and a surjection on H^{-1} (more-algebra-lemma-two-term-surjection-map-zero)

*Source reference.* Original equation 0AL4: sources/stacks-restricted/pinned-original-source.tex, source lines 326–326. Prerequisite pointer invoked at proof line(s) 2029; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 0GAG: sources/stacks-restricted/pinned-original-source.tex, source lines 669–669. Prerequisite pointer invoked at proof line(s) 2038; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original lemma 0G9I: sources/stacks-chow/tags.txt, source lines 20303–20303. Prerequisite pointer invoked at proof line(s) 2044; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 06FQ: sources/stacks-restricted/pinned-original-source.tex, source lines 426–426. Prerequisite pointer invoked at proof line(s) 2028; it names a standard fact of the same chapter that the counted argument uses, and

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every step of the counted proof built on it is retained verbatim in the delivered body.

**Lemma 0.1.** *Let $(A_1, I_1) \to (A_2, I_2)$ be as in Remark 06FQ with $A_1$ and $A_2$ Noetherian. Let $B_1$ be in (0AL4) for $(A_1, I_1)$. Let $B_2$ be the base change of $B_1$. If multiplication by $f_1 \in B_1$ on $NL^\wedge_{B_1/A_1}$ is zero in $D(B_1)$, then multiplication by the image $f_2 \in B_2$ on $NL^\wedge_{B_2/A_2}$ is zero in $D(B_2)$.*

*Proof.* By Lemma 0GAG there is a map

$$NL_{B_1/A_1} \otimes_{B_2} B_1 \to NL_{B_2/A_2}$$

which induces and isomorphism on $H^0$ and a surjection on $H^{-1}$. Thus the result by More on Algebra, Lemma 0G9I. □